

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - IV

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Queue Application:

- a. Write a program to: a. Implement a queue using an array.

```
from array import array
queue = array('i')
queue.append(10)
queue.append(20)
queue.append(30)
print("Queue:", queue)
queue.pop(0)
print("After Dequeue:", queue)
print("S009")
```

IDLE Shell 3.14.6

Edit Shell Debug Options Window Help

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jul 0 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>
= RESTART: C:/Users/mvluc/AppData/Local/Programs/Python/Python314/DS.py
Queue: array('i', [10, 20, 30])
After Dequeue: array('i', [20, 30])
S009
> |
```

- b. Simulate a simple queuing system (e.g., customer service queue).

```
import time
que = [("Adi", 2), ("Riya", 3), ("Vikas", 4), ("Sia", 6)]
print("----Booking----")
print(que)
while que:
    job, timer = que.pop(0)
    print(f"Running:{job} {timer}sec")
    time.sleep(timer)
print(que)
print("S009")
```

```
----Booking----
[('Adi', 2), ('Riya', 3), ('Vikas', 4), ('Sia', 6)]
Running:Adi 2sec
Running:Riya 3sec
Running:Vikas 4sec
Running:Sia 6sec
[]
S009
```

Curiosity Questions:

1. Where do you see queues in daily life?
2. Why is **FIFO (First In First Out)** fair in real life? Can you think of situations where it's *not* fair?
3. What will happen if you try to **enqueue more elements than the queue size**?
4. What if you **dequeue from an empty queue**?
5. Imagine you're designing a **hospital emergency system**. Should it use a simple queue or something else? (hint: priority queue).
6. If **5 customers arrive at the same time**, but the service desk can serve only 2 at once, how would you design that?